SAT Math Mistake Check

Ten original questions | Aadil Mallick | Independent practice

Use scratch paper and a calculator as needed. Record a method, not just an answer. This short check helps choose practice topics; it does not predict an SAT score. Answers start on page 3.

1. Solve for x:

$3(x - 2) = 2x + 5$

Answer: _____ Unsure? _____

2. A $100 price rises 20% and then falls 20%.

What is the final price?

Answer: _____ Unsure? _____

3. Two lines are y = 2x + 1 and y = -x + 7.

At their intersection, what is x + y?

Answer: _____ Unsure? _____

4. The equation is x^2 - 6x + 5 = 0.

What is the sum of its solutions?

Answer: _____ Unsure? _____

5. Similar triangles have side-length scale factor 3.

The smaller area is 12. What is the larger area?

Answer: _____ Unsure? _____

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Questions 6-10. Finish before checking the explanations. Mark any answer you are uncertain about.

6. A car travels at a constant 45 miles per hour for 20 minutes.

How many miles does it travel?

Answer: _____ Unsure? _____

7. Solve sqrt(x + 6) = x.

What is its only real solution?

Answer: _____ Unsure? _____

8. A fee is $12 plus $4 for each visit.

What is the total for five visits?

Answer: _____ Unsure? _____

9. The data set is 4, 6, 8, 10.

What is its mean?

Answer: _____ Unsure? _____

10. The function is f(x) = 2x^2 - 3.

What is f(-2)?

Answer: _____ Unsure? _____

Answers and explanations

An incorrect answer is a clue about the step to practice. These are original examples, not official College Board questions.

1. 11. Distribute: 3x - 6 = 2x + 5. Then x - 6 = 5 and x = 11. Check: both original sides equal 27. If you obtained 7, inspect the distribution step.

2. $96. 100 x 1.20 x 0.80 = 96. The decrease is 20% of $120, not of the original $100.

3. 7. Equating the lines gives 3x = 6, so x = 2 and y = 5. The requested quantity is x + y = 7.

4. 6. Factor as (x - 1)(x - 5). The roots are 1 and 5, whose sum is 6.

5. 108. The base and height both triple. Area scales by 3 squared, or 9. So 12 x 9 = 108.

6. 15 miles. Twenty minutes is one-third of an hour. Multiply 45 miles/hour by one-third hour.

7. 3. Squaring gives x^2 - x - 6 = 0, with candidates 3 and -2. The original fails at -2 because its right side is negative. Only 3 works.

8. $32. The fixed fee is paid once: 12 + 4 x 5 = 32.

9. 7. Add the values to get 28, then divide by the four values: 28 / 4 = 7.

10. 5. Use parentheses: 2(-2)^2 - 3 = 2 x 4 - 3 = 5.

Choose your next practice step

Do not turn the number correct into an SAT score. Instead, label each incorrect or uncertain answer, pick two topics, and practice related examples without looking at the solution.

- Concept gap: I did not understand the rule or relationship.

- Algebra error: I knew the idea but changed a sign, term, or operation.

- Reading error: I found a number but not the quantity the question requested.

- Input or unit error: I entered the expression or converted units incorrectly.

- Skipped check: I did not test whether my result satisfied the original condition.

My error log

Question / topic	My wrong step	Better method / check

Next: review one explanation, solve a new related example, and check whether the method transfers. For a scored baseline, use an appropriate official full-length practice test. If you want personal feedback, a parent or guardian can evaluate a professional tutoring option.

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